IMPLICATIONS AMONG THE TYPES OF AFFINE FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one description of a limit of rings; it lies inside the counted span and is reproduced verbatim.

Editorial note. The proof contains the author's own wording 'there exits in particular'; it lies inside the counted span and is reproduced verbatim, as the delivered text is the author's.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The delivered context span for the definition of a weakly adic ring carries a citation to [Gabber-Ramero]. Since the chapter ships no bibliography, that entry is attached to this card as bibliography.tex, copied character for character from the Stacks Project's own my.bib, so that the citation resolves without any entry being invented.

Allowed standard prerequisites (editorial).

- The types of affine formal algebraic spaces and of complete weakly admissible rings. Same chapter, the definitions of the six types appearing in the statement, used throughout. Fix the notions whose implications are asserted.
- The implications between the corresponding types of topological rings. Same chapter, Section section-weakly-adic, invoked at proof lines 2469-2474. Give the first four parts once the space is written as a formal spectrum.
- A weakly adic ring has a countable fundamental system of open ideals. Same chapter, Lemma lemma-weakly-adic at proof lines 2478-2481 and Definition definition-weakly-adic, invoked there. Gives part five.
- Affine formal algebraic spaces as colimits of thickenings of affine schemes, with McQuillan meaning a surjection onto each stage. Same chapter, Definition definition-affine-formal-algebraic-space and the defining property of the McQuillan type, used at proof lines 2485-2490. Provide the presentation and the condition verified in part six.

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Source reference. Original section 0GXD: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1588–1588. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let A be a linearly topologized ring.

(1) We say A is *weakly pre-adic*[1] if there exists an ideal $I \subset A$ such that the closure of I^n is open for all $n \geq 0$ and these closures form a fundamental system of open ideals.
(2) We say A is *weakly adic* if A is weakly pre-adic and complete[2].

Lemma 0.2. *Let X be an affine formal algebraic space over a scheme S.*

(1) *If X is Noetherian, then X is adic*.*
(2) *If X is adic*, then X is adic.*
(3) *If X is adic, then X is weakly adic.*
(4) *If X is weakly adic, then X is classical.*
(5) *If X is weakly adic, then X is countably indexed.*
(6) *If X is countably indexed, then X is McQuillan.*

Proof. Statements (1), (2), (3), and (4) follow by writing $X = \mathrm{Spf}(A)$ and where A is a weakly admissible (hence complete) linearly topologized ring and using the implications between the various types of such rings discussed in Section 0GXD.

Proof of (5). By definition there exists a weakly adic topological ring A such that $X = \operatorname{colim} \mathrm{Spec}(A/I)$ where the colimit is over the ideals of definition of A. As A is weakly adic, there exits in particular a countable fundamental system I_λ of open ideals, see Definition 0.1. Then $X = \operatorname{colim} \mathrm{Spec}(A/I_n)$ by definition of $\mathrm{Spf}(A)$. Thus X is countably indexed.

Proof of (6). Write $X = \operatorname{colim} X_n$ for some system $X_1 \to X_2 \to X_3 \to \ldots$ of thickenings of affine schemes over S. Then

$$A = \lim \Gamma(X_n, \mathcal{O}_{X_n})$$

surjects onto each $\Gamma(X_n, \mathcal{O}_{X_n})$ because the transition maps are surjections as the morphisms $X_n \to X_{n+1}$ are closed immersions. Hence X is McQuillan. □

References

[Gabber-Ramero] Gabber, Ofer and Ramero, Lorenzo. Foundations for almost ring theory–Release 7.5. arXiv preprint math/0409584, 2004.

[1] In [Gabber-Ramero] the authors say A is *c-adic*.
[2] By our conventions this includes separated.